Theoretical Foundation of PNetGimini: A Cyber-Physical Adaptive Deployment Model

Framework: Domain Decomposition with Explicit Lagrangian Integration

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1 The Global Governing Equation

The network configuration deployment process is modeled as a second-order dynamic system in a high-dimensional state space. The evolution of the configuration state vector $\mathbf{x}$ is governed by the global semi-discrete equation of motion:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}(\boldsymbol{\theta})\dot{\mathbf{x}}(t) + \mathbf{K}_{sys}\mathbf{x}(t) = \mathbf{F}_{ext}(t) + \mathbf{G}^T\boldsymbol{\lambda} \tag{1}$$

Where:

- $\mathbf{M}$ is the **Inertia Matrix**, representing the response latency and hardware constraints of network devices.
- $\mathbf{C}(\boldsymbol{\theta})$ is the **Adaptive Damping Matrix**, where $\boldsymbol{\theta} = \{T, L, A\}$ represents physical metrics (Temperature, Load, Age). It dissipates kinetic energy to prevent configuration oscillation.
- $\mathbf{K}_{sys}$ is the **Stiffness Matrix**, representing the topological coupling strength between subdomains.
- $\mathbf{F}_{ext}(t)$ is the **External Intent Force**, driving the system toward the target configuration S_{target}.
- $\mathbf{G}^T\boldsymbol{\lambda}$ represents the **Constraint Forces**, where $\boldsymbol{\lambda}$ are the Lagrangian multipliers ensuring boundary consistency.

2 Domain Decomposition and FETI Method

Using the *Finite Element Tearing and Interconnecting* (FETI) approach, the global network domain Ω is partitioned into N non-overlapping subdomains Ω_i. For each subdomain, the local Lagrangian is defined as:

$$\mathcal{L}_i(\mathbf{x}_i, \boldsymbol{\lambda}) = \frac{1}{2}\mathbf{x}_i^T\mathbf{K}_i\mathbf{x}_i - \mathbf{f}_i^T\mathbf{x}_i + \boldsymbol{\lambda}^T(\mathbf{B}_i\mathbf{x}_i) \tag{2}$$

Subject to the global interface continuity constraint:

$$\sum_{i=1}^{N} \mathbf{B}_i \mathbf{x}_i = \mathbf{0} \tag{3}$$

Where $\mathbf{B}_i$ is a signed Boolean mapping matrix that extracts interface degrees of freedom (DoF).

3 Explicit Time Integration and Stability

The system is solved using an **Explicit Central Difference Scheme**, which avoids the inversion of a global stiffness matrix, enabling massive parallelism:

$$\mathbf{x}^{(t+\Delta t)} = \mathbf{x}^{(t)} + \dot{\mathbf{x}}^{(t)} \Delta t + \frac{1}{2} \ddot{\mathbf{x}}^{(t)} \Delta t^2 \tag{4}$$

The stability is governed by the *Courant-Friedrichs-Lewy* (CFL) condition. In our model, the adaptive damping K (from PNetGimini) effectively modifies the stable time step Δt_{crit}:

$$\Delta t \leq \Delta t_{crit} \propto \sqrt{\frac{M}{K_{sys}}} \cdot \frac{1}{f(C)} \tag{5}$$

This mathematically justifies the **"Safe Crawling Mode"** when physical environment metrics degrade.

4 Boundary Coordination via ADMM

The update of Lagrangian multipliers $\boldsymbol{\lambda}$ (representing boundary pressure) follows the *Alternating Direction Method of Multipliers* (ADMM) logic:

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \sum_{i=1}^{N} \mathbf{B}_i \mathbf{x}_i^{(k+1)} \tag{6}$$

Where ρ is the augmentation parameter. This iterative process ensures that the configuration convergence residual ϵ satisfies:

$$R = \| \sum \mathbf{B}_i \mathbf{x}_i \| < \epsilon \tag{7}$$